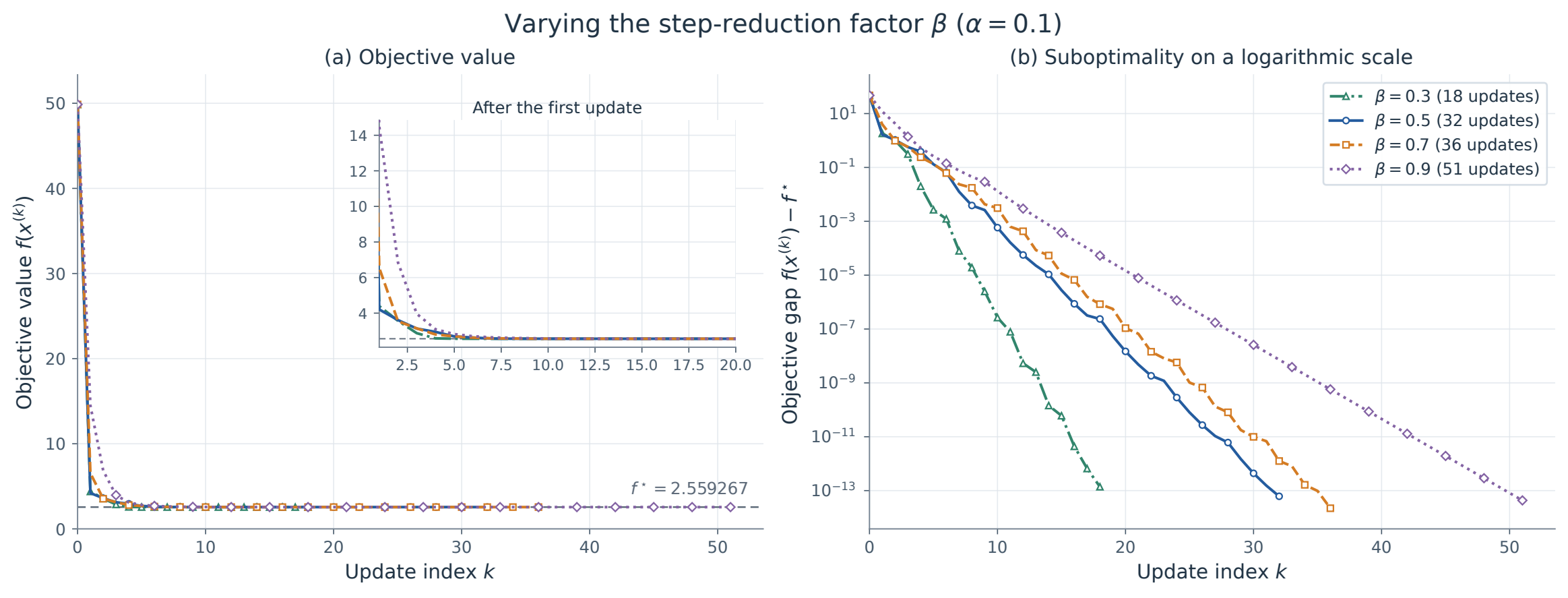
Varying the step-reduction factor β ($\alpha = 0.1$)
(a) Objective value
After the first update
Objective value $f(x^{(k)})$
Update index k
$f^\star = 2.559267$
(b) Suboptimality on a logarithmic scale
Objective gap $f(x^{(k)}) - f^\star$
Update index k
$\beta = 0.3$ (18 updates)
$\beta = 0.5$ (32 updates)
$\beta = 0.7$ (36 updates)
$\beta = 0.9$ (51 updates)